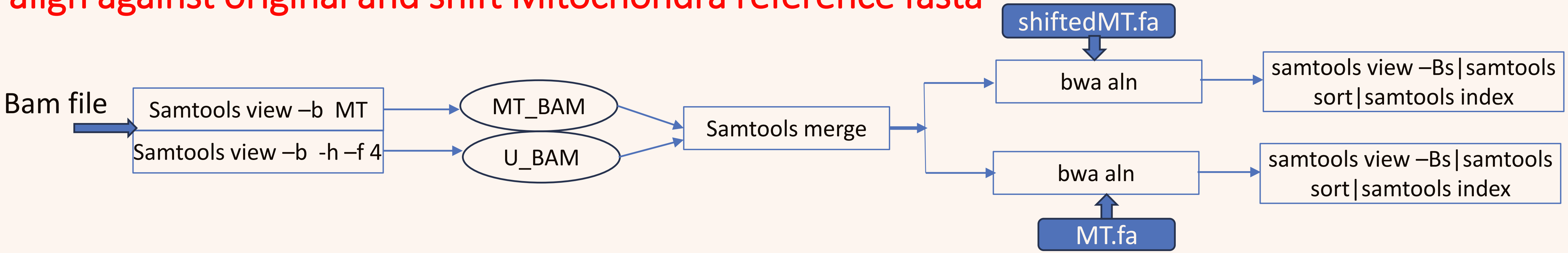
User Manual of a Nextflow Pipeline for Mitochondria Variant Calling

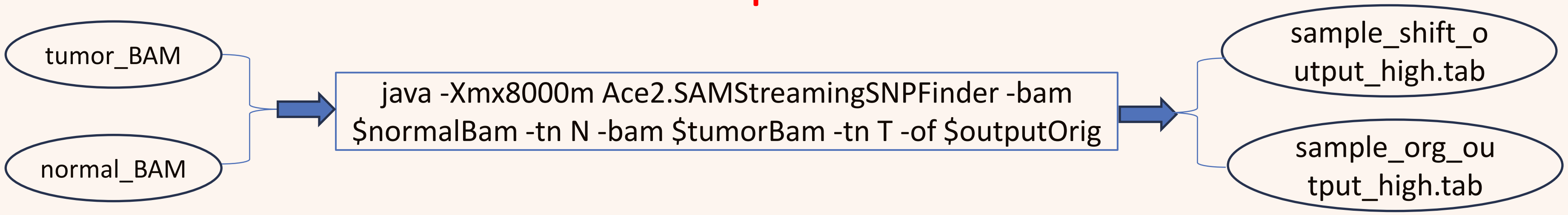
Wenchao Zhang

This Current Mitochondra pipeline include three function parts

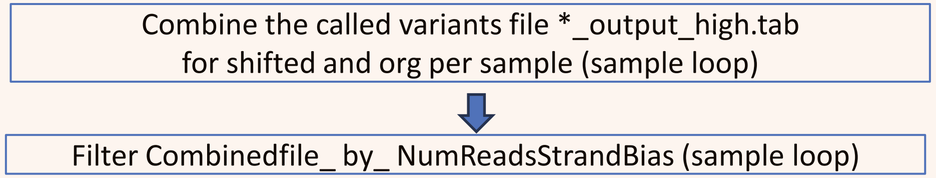
Function part1- Extract and filter bam into mitochondria and unaligned bam and use bwa to align against original and shift Mitochondra reference fasta



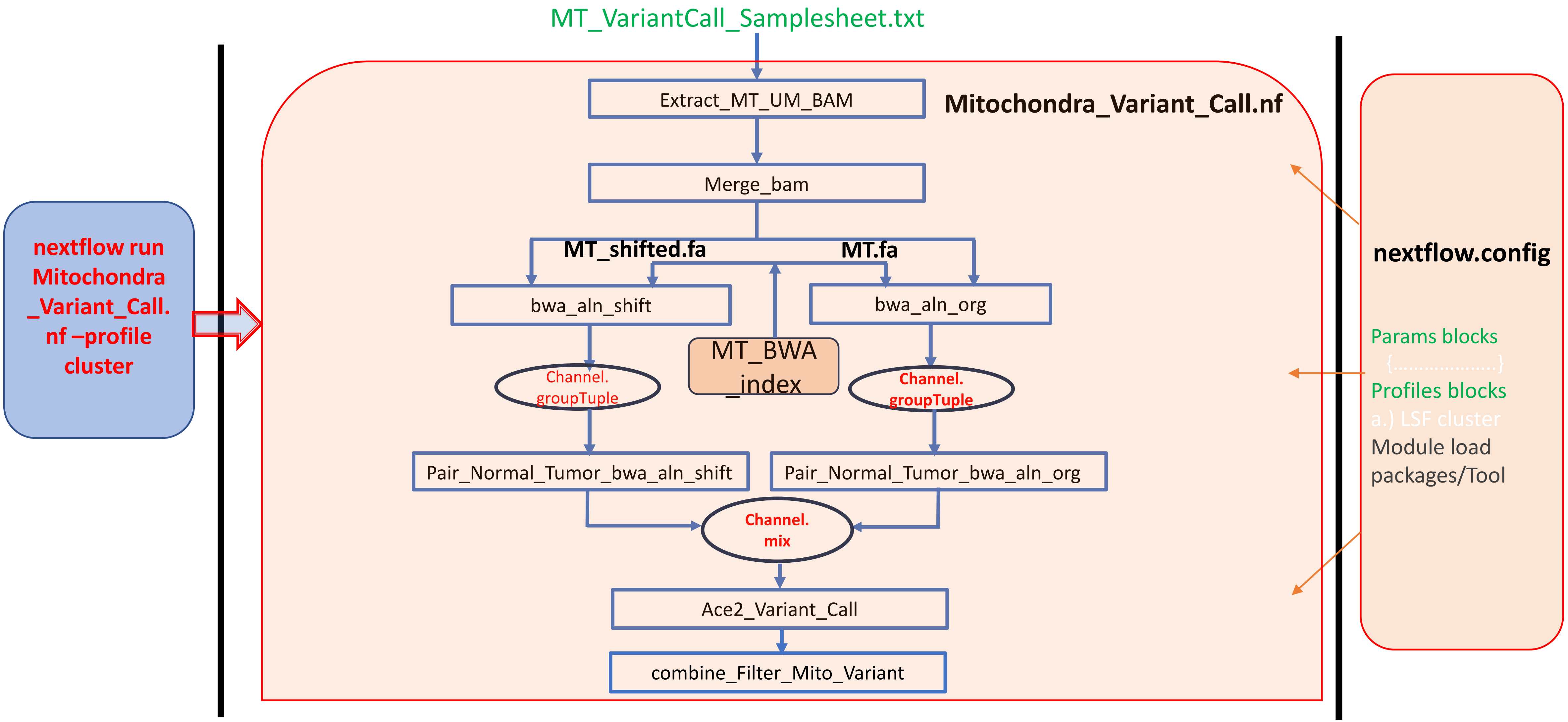
Function part2- Call the variant based on the paired Mitochondra tumor and normal BAM pair



Function part3- Combine the two type of called variants and do some filtering



Architecture and Workflow of Nextflow pipeline- **Mitochondra_Variant_Call.nf**



Samplesheet

- $ cat MT_VariantCall_Samplesheet.txt

```
Sample_062428   Normal  /research/groups/cab/projects/automapper/common/wzhang42/RB1_iPSC/WU-RB1_iPSC-WGS/SJNORM062428_C1/20242082l/SJNORM062428_C1.marked_dup.recal.bam
Sample_062428   Tumor   /research/groups/cab/projects/automapper/common/wzhang42/RB1_iPSC/WU-RB1_iPSC-WGS/SJRB062428_O1/20242O827/SJRB062428_O1.marked_dup.recal.bam
Sample_062428_100M      Normal  /research/groups/cab/projects/automapper/common/wzhang42/RB1_iPSC/WU-RB1_iPSC-WGS/SJNORM062428_C1_100M/20242O823/SJNORM062428_C1_100M.marked_dup.recal.bam
Sample_062428_100M      Tumor   /research/groups/cab/projects/automapper/common/wzhang42/RB1_iPSC/WU-RB1_iPSC-WGS/SJRB062428_O1_100M/20242O830/SJRB062428_O1_100M.marked_dup.recal.bam
```

nextflow.config

Input

Output

```
params {
    h =null
    help =null
    Mitochondra_samplelist  = "./MT_VariantCall_Samplesheet.txt"
    //a txt file that recording the Mitochondra Normal and Tumor aligned BAMs. Each row has three columns that are separated by TAB and correspond to Sample_Name, Normal/Turmor, Normal_BAM/
Tumor_BAM. Two rows are corresponds to one normal_tumor pair.  So, the samplesheet should be as " Sample1, normal1, norm1.bam; Sample1, tumor1, tumor1.bam; Sample2, normal2, norm2.bam; Sample2,
tumor2, tumor2.bam;

    project = "MT_VariantCall_Test"
    //A project atlas for the current Mitochondra somatic variant calling

    Ref_MT = "/research/groups/cab/projects/automapper/common/wzhang42/Mitochondra_Pipeline/MT_bwa/MT.fa"
    // Mitochondra Original reference Fasta file
    Ref_Shifted_MT = "/research/groups/cab/projects/automapper/common/wzhang42/Mitochondra_Pipeline/MT_bwa/shiftedMT.fa"
    // Mitochondra shifted reference Fasta file

    Select_Variant_Call = "Y"
    // Whether select to call Mitochondra somatic variant calling

    Select_Variant_Combine_Filter = "Y"
    // Whether select to combine and filter the called variants for downstream annotation

    Python_SCript_Path="/research/groups/cab/projects/automapper/common/wzhang42/Mitochondra_Pipeline/Python_Scripts"
    // Python script in this pipeline, such as Combine_FilterbyNumberReadStrandBias.py

    Java_LIB_Path= "/research/groups/cab/projects/automapper/common/wzhang42/Mitochondra_Pipeline/Java_LIBs"
    // The Path for Java lib files (.Jar) that will be used in Ace_Variant_Call

    outdir= "." // "/scratch_space/wzhang42"
    // Output dir for storing the intermediate publishedDir

    TMP_DIR="/scratch_space/wzhang42/tmp"

}
```

bsub -P Mitocondra_test -q rhel8_standard -M 50000 -e err%J.err -o out%J.out -J Mitocondra_test "module load nextflow/21.10.5 && nextflow run ./Mitochondra_Variant_Call.nf -profile cluster"

```
(base) [wzhang42@splprhpc09 Nextflow_Package]$ cat out212230166.out
Loading nextflow/21.10.5
  Loading requirement: java/17.0.1
N E X T F L O W  ~  version 21.10.6
Launching `./Mitochondra_Variant_Call.nf` [deadly_pare] - revision: e3dd211e87

Welocme to run Nextflow Pipeline Mitochondra_Variant_Call.nf
Your configuration are the following:
  project                         : MT_VariantCall_Test
  Mitochondra_samplelist          : ./MT_VariantCall_Samplesheet.txt
  Ref_MT                          : /research/groups/cab/projects/automapper/common/wzhang42/Mitochondra_Pipeline/MT_bwa/MT.fa
  Ref_Shifted_MT                  : /research/groups/cab/projects/automapper/common/wzhang42/Mitochondra_Pipeline/MT_bwa/shiftedMT.fa
  Select_Variant_Call             : Y
  Select_Variant_Combine_Filter   : Y
  Python_SCript_Path              : /research/groups/cab/projects/automapper/common/wzhang42/Mitochondra_Pipeline/Python_Scripts
  Java_LIB_Path                   : /research/groups/cab/projects/automapper/common/wzhang42/Mitochondra_Pipeline/Java_LIBs
  outdir                          : .
```

```
executor >  lsf (42)
[2e/5fe04c] process > Extract_MT_UM_BAM (1)          [100%] 6 of 6 ✔
[08/ae35ca] process > Merge_bam (6)                  [100%] 6 of 6 ✔
[2c/b4db88] process > bwa_aln_shift (6)              [100%] 6 of 6 ✔
[78/f71b76] process > Pair_Normal_Tumor_bwa_aln_s... [100%] 3 of 3 ✔
[43/472d39] process > bwa_aln_org (6)                [100%] 6 of 6 ✔
[7b/08a881] process > Pair_Normal_Tumor_bwa_aln_o... [100%] 3 of 3 ✔
[fa/32d2c8] process > Ace2_Variant_Call (6)          [100%] 6 of 6 ✔
[b6/47d0ca] process > Combine_Filter_Mito_Variant... [100%] 3 of 3 ✔
[71/f0e188] process > Annotate_Mito_Variant (3)      [100%] 3 of 3 ✔
Completed at: 16-Jan-2024 13:01:09
Duration    : 39m 4s
CPU hours   : 7.2
Succeeded   : 42
```

$ ls MT_VariantCall_Test/

Ace2_Variant_Call bwa_aln_org bwa_aln_shift Combine_Filter_Mito_Variant Extract_MT_UM_BAM Merge_bam

$ ls Ace2_Variant_Call/Sample_062428

Sample_062428_org_output_high.tab Sample_062428_shift_output_high.tab

- $ cat Ace2_Variant_Call/Sample_062428/Sample_062428_org_output_high.tab|less

```
NormalSample    TumorSample     Name    Chr     Pos     Type    Size    Coverage        Percent_alternative_allele      Chr_Allele      Alternative_Allele      Score   Text    unique_alternative_ids  reference_normal_count  reference_tum
or_count  alternative_normal_count        alternative_tumor_count count_ref_normal_fwd    count_ref_normal_rev    count_ref_tumor_fwd     count_ref_tumor_rev     count_var_normal_fwd    count_var_normal_rev    count_var_tumor_fwd     co
unt_var_tumor_rev       alternative_fwd_count   alternative_rev_count   alternative_bidirectional_confirmation  broad_coverage  broad_reference_normal_count    broad_reference_tumor_count     broad_alternative_normal_count  broad_alternati
ve_tumor_count  unique_alt_read_starts  unique_alt_read_starts_fwd      unique_alt_read_starts_rev      avg_mapq_alternative    somatic_or_germline     loh_flag        alt_to_ref_ratio_normal alt_to_ref_ratio_tumor  alt_to_ref_ratio_dif
f_normal_tumor  strand_skew
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.7  chrM    7       SNP     1       1377    0.001   A       T       0.500   GATCAC[A/T]GGTCTATCACCCTATTAACC 1       864     512     1       0       425
    439     216     296     0       1       0       0       0       1       0       1465    915     549     1       0       1       0       1       37              0.001   0.000   -0.001
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.14 chrM    14      SNP     1       2696    0.000   T       C       0.500   GATCACAGGTCTA[T/C]CACCCTATTAACCACTCACG  1       1754    941     1       0
    856     898     431     510     0       1       0       0       0       1       0       2872    1858    1012    2       0       1       0       1       37              0.001   0.000   -0.001
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.16 chrM    16      SNP     1       3014    0.000   A       G       0.090   GATCACAGGTCTATC[A/G]CCCTATTAACCACTCACGGG        1       1983    1030    1
    0       1010    973     489     541     1       0       0       0       1       0       3211    2079    1130    2       0       1       1       0       37              0.001   0.000   -0.001
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.18 chrM    18      SNP     1       3366    0.000   C       A       0.500   GATCACAGGTCTATCAC[C/A]CTATTAACCACTCACGGGAG      1       2218    1147    0
    1       1117    1101    537     610     0       0       1       0       1       0       0       3658    2384    1273    0       1       1       1       0       37              0.000   0.001   0.001
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.19 chrM    19      SNP     1       3550    0.000   C       T       0.500   GATCACAGGTCTATCACC[C/T]TATTAACCACTCACGGGAGC     1       2343    1206    0
    1       1171    1172    576     630     0       0       0       1       0       1       0       3875    2511    1363    0       1       1       0       1       37              0.000   0.001   0.001
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.20 chrM    20      SNP     1       3734    0.000   T       A       0.090   GATCACAGGTCTATCACCC[T/A]ATTAACCACTCACGGGAGCT    1       2458    1274    1
    0       1239    1219    619     655     1       0       0       0       1       0       0       4087    2642    1444    1       0       1       1       0       37              0.000   0.000   -0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.23 chrM    23      SNP     1       4299    0.000   T       G       0.090   TCACAGGTCTATCACCCTAT[T/G]AACCACTCACGGGAGCTCTC   1       2872    1426    0
    1       1433    1439    701     725     0       0       0       1       0       1       0       4661    3034    1624    0       3       1       0       1       37              0.000   0.002   0.002
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.25 chrM    25      SNP     1       4676    0.000   A       G       0.999   ACAGGTCTATCACCCTATTA[A/G]CCACTCACGGGAGCTCTCCA   2       3087    1587    2
    0       1555    1532    778     809     2       0       0       0       2       0       0       5072    3270    1800    2       0       2       2       0       37              0.001   0.000   -0.001
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.26 chrM    26      SNP     1       4872    0.000   C       A       0.500   CAGGTCTATCACCCTATTAA[C/A]CACTCACGGGAGCTCTCCAT   1       3198    1673    1
    0       1620    1578    816     857     1       0       0       0       1       0       0       5272    3387    1883    2       0       1       1       0       37              0.001   0.000   -0.001
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.27 chrM    27      SNP     1       4983    0.000   C       A       0.999   AGGTCTATCACCCTATTAAC[C/A]ACTCACGGGAGCTCTCCATG   2       3286    1695    1
    1       1703    1583    854     841     1       0       1       0       2       0       0       5466    3513    1951    1       1       2       2       0       37              0.000   0.001   0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.29 chrM    29      SNP     1       5242    0.000   C       G       0.500   GTCTATCACCCTATTAACCA[C/G]TCACGGGAGCTCTCCATGCA   1       3478    1762    0
    1       1809    1669    880     882     0       0       0       1       0       1       0       5827    3766    2060    0       1       1       0       1       37              0.000   0.000   0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.30 chrM    30      SNP     1       5449    0.000   T       G       0.090   TCTATCACCCTATTAACCAC[T/G]CACGGGAGCTCTCCATGCAT   1       3587    1861    0
    1       1859    1728    915     946     0       0       1       0       1       0       0       6030    3896    2133    0       1       1       1       0       37              0.000   0.000   0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.31 chrM    31      SNP     1       5672    0.000   C       T       0.999   CTATCACCCTATTAACCACT[C/T]ACGGGAGCTCTCCATGCATT   2       3721    1949    1
    1       1908    1813    947     1002    0       1       1       0       1       1       1       6261    4051    2207    1       2       2       1       1       37              0.000   0.001   0.001
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.32 chrM    32      SNP     1       5835    0.000   A       G       0.909   TATCACCCTATTAACCACTC[A/G]CGGGAGCTCTCCATGCATTT   2       3836    1997    2
    0       1999    1837    971     1026    2       0       0       0       2       0       0       6444    4187    2255    2       0       2       2       0       37              0.000   0.000   -0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.33 chrM    33      SNP     1       5994    0.000   C       T       0.999   ATCACCCTATTAACCACTCA[C/T]GGGAGCTCTCCATGCATTTG   2       3933    2059    2
    0       2056    1877    1003    1056    2       0       0       0       2       0       0       6687    4360    2322    3       2       2       2       0       37              0.001   0.001   0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.34 chrM    34      deletion        1       6367    0.000   G               0.090   1       4181    2185    1       0       2157    2024    1060
  1125  1       0       0       0       1       0       0       6991    4574    2416    1       0       1       1       0       37              0.000   0.000   -0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.36 chrM    36      SNP     1       6796    0.000   G       A       0.999   ACCCTATTAACCACTCACGG[G/A]AGCTCTCCATGCATTTGGTA   2       4440    2354    2
    0       2285    2155    1133    1221    1       1       0       0       1       1       1       7445    4843    2598    3       1       2       1       1       37              0.001   0.000   -0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.37 chrM    37      SNP     1       6994    0.000   A       G       0.500   CCCTATTAACCACTCACGGG[A/G]GCTCTCCATGCATTTGGTAT   1       4560    2432    0
    1       2352    2208    1172    1260    0       0       1       0       1       0       0       7639    4961    2676    1       1       1       1       0       37              0.000   0.000   0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.38 chrM    38      SNP     1       7306    0.000   G       A       0.909   CCTATTAACCACTCACGGGA[G/A]CTCTCCATGCATTTGGTATT   2       4743    2559    2
    0       2412    2331    1212    1347    0       2       0       0       0       2       0       7826    5089    2735    2       0       2       0       2       37              0.000   0.000   -0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.39 chrM    39      SNP     1       7636    0.000   C       A       0.999   CTATTAACCACTCACGGGAG[C/A]TCTCCATGCATTTGGTATTT   2       4951    2681    2
    0       2520    2431    1262    1419    1       1       0       0       1       1       1       8114    5271    2841    2       0       2       1       1       37              0.000   0.000   -0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.40 chrM    40      SNP     1       7882    0.000   T       A       0.500   TATTAACCACTCACGGGAGC[T/A]CTCCATGCATTTGGTATTTT   1       5311    2569    0
    1       2668    2643    1280    1289    0       0       1       0       1       0       0       8535    5645    2887    1       2       1       1       0       37              0.000   0.001   0.001
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.43 chrM    43      deletion        1       8553    0.000   C               0.500   1       5789    2760    1       0       2944    2845    1353
  1407  0       1       0       0       0       1       0       9372    6171    3200    1       0       1       0       1       37              0.000   0.000   -0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.42 chrM    42      SNP     1       8330    0.000   T       A       1.000   TTAACCACTCACGGGAGCTC[T/A]CCATGCATTTGGTATTTTCG   3       5612    2712    3
    0       2858    2754    1332    1380    1       2       0       0       1       2       1       9102    5989    3109    4       0       3       1       2       37              0.001   0.000   -0.001  0.330
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.43 chrM    43      SNP     1       8552    0.000   C       T       0.999   TAACCACTCACGGGAGCTCT[C/T]CATGCATTTGGTATTTTCGT   2       5789    2760    2
    0       2944    2845    1353    1407    1       1       0       0       1       1       1       9373    6171    3200    2       0       2       1       1       37              0.000   0.000   -0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.44 chrM    44      SNP     1       8763    0.000   C       A       0.500   AACCACTCACGGGAGCTCTC[C/A]ATGCATTTGGTATTTTCGTC   1       5972    2789    0
    1       3048    2924    1365    1424    0       0       0       1       0       1       0       9640    6363    3274    0       3       1       0       1       25              0.000   0.001   0.001
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.45 chrM    45      SNP     1       8926    0.000   A       T       0.500   ACCACTCACGGGAGCTCTCC[A/T]TGCATTTGGTATTTTCGTCT   1       6116    2809    1
    0       3134    2982    1370    1439    0       1       0       0       0       1       0       9820    6496    3321    3       0       1       0       1       37              0.000   0.000   -0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.46 chrM    46      deletion        1       9170    0.000   T               0.500   1       6361    2808    1       0       3228    3133    1385
  1423  1       0       0       0       1       0       0       10097   6691    3405    1       0       1       1       0       37              0.000   0.000   -0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.47 chrM    47      insertion       1       9380    0.000           C       0.500   1       6513    2862    1       0       3307    3206    1425
  1437  0       1       0       0       0       1       0       10340   6821    3518    1       0       1       0       1       37              0.000   0.000   -0.000
Sample_062428_org_normal_aln_sort.bam   Sample_062428_org_tumor_aln_sort.bam    chrM.47 chrM    47      SNP     1       9380    0.000   G       A       1.000   CACTCACGGGAGCTCTCCAT[G/A]CATTTGGTATTTTCGTCTGG   4       6513    2862    3
:
```